



GM12878 Hi-C

Download the .fastq files with SRATools:

- Download SRATools from the ncbi repository: <https://github.com/ncbi/sra-tools>, following the guide in <https://github.com/ncbi/sra-tools/wiki/02.-Installing-SRA-Toolkit>
- Set up the file with accessions at
`/scratch/users/kalyanna/GM12878_hic/GM128787_Rao2014_SRA.txt`
- Run a bash script for downloads:

```
#!/bin/bash

accession_file="/scratch/users/kalyanna/GM12878_hic/GM128787_Rao2014_SRA.txt"

output_dir="/scratch/users/kalyanna/GM12878_hic/sra_downloads"

while IFS= read -r accession
do
    echo "Sending sbatch script for $accession"
    sbatch ./sbatch.slurm $accession $output_dir
done < "$accession_file"
```

Using the following slurm script:

```
#!/bin/bash
#SBATCH -c 1
#SBATCH --mem=8G
#SBATCH --partition=normal,altemose,owners
#SBATCH --time=48:00:00
#SBATCH --chdir=/scratch/users/kalyanna/GM12878_hic/
#SBATCH --export=all
#SBATCH --requeue
#SBATCH --output=/scratch/users/kalyanna/GM12878_hic/sbatch_logs/dow
```

```
source activate hic
module load biology sra-tools

echo "Processing $1"

prefetch -O "$2" --max-size 1000000000000 $1
fastq-dump --split-3 -O "$2" "$2/$1/$1.sra"
```

- The SRA files are stored at / `scratch/users/kalyanna/GM12878_hic/sra_downloads`

Set up the configuration file

From the paper: "DNA was digested with 100 units of Mbol, and the ends of restriction fragments were labeled using biotinylated nucleotides and ligated in a small volume. After reversal of crosslinks, ligated DNA was purified and sheared to a length of ~400 bp, at which point ligation junctions were pulled down with streptavidin beads and prepped for Illumina sequencing. "

The configuration file is at `/scratch/users/kalyanna/GM12878_hic/config-hic_gm12878.txt`

Map replicates separately

The rawdata from different replicates (Read_1.fastq and Read_2.fastq) is assigned to different folders in `rawdata/${raplicate_name}/${raplicate_name}`. The `rawdata/${raplicate_name}/` folder is passed to the HiC-Pro for mapping and qc. The script `send_sbatch_mapping_qc.sh` allows iteratively send slurm script for all replicates:

```
#!/bin/bash

working_dir="/scratch/users/kalyanna/GM12878_hic/rawdata"
cd "$working_dir"

for dir in */; do
    dir=${dir%/}
    subdir="$working_dir/$dir"
    output="/scratch/users/kalyanna/GM12878_hic/results_$dir"

    echo "Sending $subdir"
```

```
sbatch /scratch/users/kalyanna/GM12878_hic/sbatch_scripts/GM12878_hic/
done
```

The slurm script is the following:

```
#!/bin/bash
#SBATCH -c 32
#SBATCH --mem=240G
#SBATCH --partition=owners
#SBATCH --time=36:00:00
#SBATCH --chdir=/scratch/users/kalyanna/GM12878_hic/
#SBATCH --export=all
#SBATCH --requeue
#SBATCH --output=/scratch/users/kalyanna/GM12878_hic/sbatch_logs/sbatch_%.x.log

source activate hic

echo "Processing $1"

/scratch/users/kalyanna/HiC-Pro_3.1.0/bin/HiC-Pro \
-i "$1" \
-c /scratch/users/kalyanna/GM12878_hic/config-hic_gm12878.txt \
-o "$2" \
-s mapping -s proc_hic -s quality_checks
```

This code creates multiple folders named `results_${replicate_name}`, where it stores bowtie2 results and hic results.

Transfer all `.validPairs` files from `hic_results` into `merged_sample` folder

The results for separate replicates are available at separate subfolders, named `results_${replicate_name}`. Now, when `.validPairs` and `.bam` files are ready, we need to construct the matrices for hic interactions.

- Pull all results to a single **merged_samples** folder stored at `/scratch/users/kalyanna/GM12878_hic/results_SRR1658573/hic_results/data/merged_samples`
- Iterate through folders with `results_SRR*/hic_results/data/SRR*/*.validPairs`

```
#!/bin/bash

cd /scratch/users/kalyanna/GM12878_hic

for dir in results_*;
```

```
do
    dir=${dir#*_}

    copy_from_dir="/scratch/users/kalyanna/GM12878_hic/results_$d
    copy_to_dir="/scratch/users/kalyanna/GM12878_hic/results_SRR1
    echo "Retrieving from $copy_from_dir to $copy_to_dir"
    cp $copy_from_dir $copy_to_dir
done
```

Build maps

After the .validPairs are transferred to the merged_samples folder, we need to merge the sample in .allValidPairs file. Next, run the map builder and normalisation. The contact matrices are build at resolutions of [10000 20000 40000 80000 160000 320000 640000]

```
#!/bin/bash
#SBATCH -c 32
#SBATCH --mem=240G
#SBATCH --partition=owners
#SBATCH --time=36:00:00
#SBATCH --chdir=/scratch/users/kalyanna/GM12878_hic/
#SBATCH --export=all
#SBATCH --requeue
#SBATCH --output=/scratch/users/kalyanna/GM12878_hic/sbatch_logs/sba

source activate hic

/scratch/users/kalyanna/HiC-Pro_3.1.0/bin/HiC-Pro \
-c /scratch/users/kalyanna/GM12878_hic/config-hic_gm12878.txt \
-i /scratch/users/kalyanna/GM12878_hic/results_SRR1658573/hic_result
-o /scratch/users/kalyanna/GM12878_hic/results_SRR1658573/ \
-s merge_persample -s build_contact_maps -s ice_norm
```

Convert .matrix files to .cool files, .cool to .npz

I am using hicConvert to convert matrices to .cool format. Then, I convert .cool to npz for easier visualisation:

```
resolutions=(10000 20000 40000 80000 160000 320000 640000)

output_dir_cool="/scratch/users/kalyanna/GM12878_hic/cool"

```

```

output_dir_npz="/scratch/users/kalyanna/GM12878_hic/npz"

mkdir -p $output_dir_cool
mkdir -p $output_dir_npz

for res in "${resolutions[@]}";
do
    echo "Resolution is $res"
    echo "Matrix is at /scratch/users/kalyanna/GM12878_hic/results_S"
    echo "Bed is at /scratch/users/kalyanna/GM12878_hic/results_SRR1"
    echo "Outputing to the $output_dir_cool/merged_sample_GM12878_{$"
    hicConvertFormat -m /scratch/users/kalyanna/GM12878_hic/results_
    --bedFileHicpro scratch/users/kalyanna/GM12878_hic/results_SRR16
    --inputFormat hicpro \
    --outputFormat cool \
    -o $output_dir_cool/merged_sample_GM12878_{$res}.cool
done

cd $output_dir_npz

for res in "${resolutions[@]}";
do
    mkdir -p npz_$res
    echo "Getting $output_dir_cool/merged_sample_GM12878_{$res}.cool"
    echo "Outputing to npz_$res"
    echo "The resolution is $res"
    python /scratch/users/kalyanna/C.Origami/src/corigami/preprocess
    $output_dir_cool/merged_sample_GM12878_{$res}.cool \
    npz_$res \
    -r $res \
    --no-balance
done

```

Final results

Npz matrices are stored at : `/scratch/users/kalyanna/GM12878_hic/npz/npz_$res`

Cool files are stored at: `/scratch/users/kalyanna/GM12878_hic/cool/`